

Uddipan Basu Bir

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PROFESSIONAL SUMMARY

I am in the final phase of my M.Sc. in Data Science with a specialization in AI and ML, having submitted my thesis and currently awaiting the results. I bring close to 5 years of experience, including 3.5 years in full-time roles and close to 2 years as a working student, across Data Analytics, Data Engineering, and Applied AI. Through my work at Siemens, Schaeffler, and Tata Consultancy Services, I have built end-to-end AI/ML solutions, production-level RAG systems, NLP/LLM applications, ETL workflows, dashboards, and automation solutions. My technical background includes Python, SQL, PySpark, Power BI, LangChain, LangGraph, LlamaIndex, and cloud platforms such as Azure, AWS, and GCP, along with CI/CD and monitoring tools. My broader interests lie in applied AI, machine learning, multimodal systems, and scalable data-driven solutions.

EXPERIENCE

Munich Music Labs, TUM Germany

Munich, Germany

Research Team Member

Nov 2025 – Present

- Developing a LLM/foundation-model-driven pipeline to transcribe audio into symbolic music.

Siemens AG

Erlangen, Germany

Working Student – Applied AI Engineer

May 2025 – Present

- Built an AI reporting pipeline that unified departmental data into a single reporting layer and generated automated executive PDF/PPT summaries.
- Built an NLP-to-SPARQL interface for natural-language querying of GraphDB.

Tools & Technologies: Snowflake, dbt, Microsoft Copilot Studio, Azure Machine Learning, Azure OpenAI, LangChain, PyTorch, Azure AI Search, Helix AI, Postman Collection.

Schaeffler AG

Nürnberg, Germany

Working Student – Data Analyst

Jun 2024 – Apr 2025

- Built interactive Power BI dashboards for BOM and financial reporting, including data cleaning, transformation, and data model design to support accurate visual analysis.
- Automated SAP report extraction and SharePoint-based reporting workflows using Power Automate to support downstream Power BI refresh.

Tools & Technologies: Power BI, Power Apps, Power Automate, SAP, Dataverse, Microsoft Power Platform.

Tata Consultancy Services

Kolkata, India

Data Engineer

Sep 2018 – Jun 2022

Sep 2021 – Jun 2022

- Built and maintained 10+ ETL workflows in Ab-Initio using Autosys scheduling, improving data delivery SLA by 30%.
- Optimized ETL processes by migrating logic from Ab-Initio to PySpark, reducing runtime by 20%.

Data Analyst

Sep 2018 – Aug 2021

- Developed Power BI dashboards and reports (MRR, ARR, and P&L), enabling faster data-driven decisions.
- Automated data quality checks and implemented cleaning scripts, reducing manual effort by 20%.

Tools & Technologies: Kafka, Airflow, GitHub Enterprise, Python, Unix, Autosys, Ab-Initio, SQL Indexing, Window Functions, CTEs, SQL Procedures, ServiceNow, Power BI, Power Apps, SQL.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

Erlangen, Germany

M.Sc. in Data Science; Specialization in Artificial Intelligence & Machine Learning

Oct 2022 – Present

- **Master Thesis:** Comparative analysis of vision-language models for OCR-based structured JSON extraction from document images.
- **Master Seminar Project (REHAU & FAU):** Forecasting and anomaly detection in time series data for polymer extrusion process. [GitHub](#)

Maulana Abul Kalam Azad University of Technology

Kolkata, India

Bachelor of Technology in Computer Science & Engineering

Jun 2014 – Jul 2018

- **Bachelor Thesis:** Emotion Detection Through Facial Expressions. [GitHub](#)

SKILLS

Programming Languages: Python, SQL, Unix Shell Scripting, TypeScript

Machine Learning & Deep Learning: PyTorch, PyTorch Lightning, Scikit-learn, XGBoost, statsmodels

NLP / LLM Tooling: Hugging Face Transformers, LangChain, LangGraph, Groq Cloud, SWIFT, LLaMA-Factory, NLTK, spaCy, vLLM, LangFlow, n8n

Vector Databases: OpenSearch, Qdrant, FAISS, Chroma

Data Engineering & Orchestration: PySpark, Airflow, Autosys, Ab-Initio

Cloud & HPC: Azure, AWS, HPC, SLURM

Experiment Tracking & HPO: Opik, MLflow, Weights & Biases, TensorBoard, Optuna

Web & API Frameworks: FastAPI, Flask, Node.js, Streamlit

BI / Automation / Monitoring: Power BI, Power Automate, Power Apps, Prometheus, Grafana, Docker, Kubernetes

PROJECTS

Master Thesis: From Pixels to Structure | [GitHub](#)

- Fine-tuning and comparative analysis of multimodal and OCR-based vision-language models using PEFT methods such as LoRA, QLoRA, and Unsloth for structured JSON extraction from handwritten and printed document images.
- **Tools & Technologies:** Vision-Language Models, OCR, PEFT, LoRA, QLoRA, Unsloth, Structured JSON Extraction.

Multi-modal, multi-view, multi-task Industrial Quality Inspection | [GitHub](#)

- Developed a multi-modal, multi-view, multi-task deep learning pipeline for classification and quality monitoring in industrial manufacturing.
- **Tools & Technologies:** Deep Learning, Multi-Modal Learning, Multi-View Learning, Multi-Task Learning, Computer Vision, Industrial Quality Monitoring.

Self-/Semi-Supervised Industrial Inspection Study | [GitHub](#)

- Built and compared self-supervised, semi-supervised, and combined deep learning approaches for image classification in industrial visual inspection of electrical motor manufacturing.
- **Tools & Technologies:** Self-Supervised Learning, Semi-Supervised Learning, Deep Learning, Image Classification, Computer Vision, Industrial Visual Inspection.

Local LLM-based RAG System for Personal Documents | [GitHub](#)

- Built a local, privacy-preserving RAG system for German bureaucracy letters using Streamlit, Ollama, OpenSearch hybrid retrieval, Redis caching, OCR-based ingestion, and Prometheus/Grafana observability, fully deployed via Docker.
- **Tools & Technologies:** Streamlit, Ollama, OpenSearch, Hybrid Search, Redis, Prometheus, Grafana, Docker, OCR.

Accident Severity Prediction from Sentiment Embeddings | [GitHub](#)

- Built a sentiment analysis pipeline on news reports to predict accident severity by comparing GloVe, Word2Vec, FastText, and Sentence Transformers embeddings with a Random Forest classifier.
- **Tools & Technologies:** Sentiment Analysis, GloVe, Word2Vec, FastText, Sentence Transformers, Random Forest, Text Embeddings.

HACKATHON

Agentic Fraud Detection Hackathon – [Reply](#) | [GitHub](#)

- Built three independent agentic fraud-detection pipelines using LangChain, Groq, and Anthropic SDK for a multi-modal financial dataset spanning transactions, SMS, email, and audio, with dual-agent review patterns and Langfuse observability tracing.
- Designed a cost-optimized three-tier LLM cascade (rule-based screen → Claude Haiku → Claude Sonnet escalation), reducing inference cost to ~5% of a naive approach, with local audio transcription via faster-whisper and parallel batch processing.

LANGUAGES

English: Fluent **German:** Intermediate **Bengali:** Native **Hindi:** Fluent